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2.1 Introduction

This chapter presents a review of the literature relevant to the development of our proposed system. It describes the core concept of enterprise architecture (EA) with the emphasis being put on its potential benefits and possible challenges. The discussion also delves into the different types of EA frameworks used to manage the architecture. A relevant example of it established in the higher education setting is also presented. The chapter further explores various sub-systems which make up the enterprise architecture of a university, such as student information, finance, learning and HR systems. Additionally, it takes a closer look at the current and past systems used at UTHM and their benefits. The chapter also presents the technology commonly used when designing such systems.

2.2 Enterprise Architecture (EA)

2.2.1 Definition and Concepts

Enterprise Architecture (EA) is a strategic procedure that helps an organisation manage complex operations and align their technologies, processes and resources with their business goals. Instead of allowing different systems and departments to operate separately, EA provides an overall blueprint showing how business functions, processes, information and technology platforms are interdependent (Kaisler et al., 2005). In today's rapidly changing business environment, organisations often struggle with isolated systems and unclear relationships between business and IT assets, which makes large-scale conversion work difficult and inefficient (Bloch et al., 2012). EA solves this problem by providing an integrated view of the enterprise and guiding the development of current and future architecture, so as to achieve more coordinated decision-making and more effective organisational changes (Niemi E., 2020).

There are two common EA frameworks are used to help organizations structure and manage their architecture:

1. Zachman Framework

Zachman Framework acts like a classification chart or a structured table for organizing all the important descriptions of an enterprise. It uses a matrix with six basic questions, “What, How, Where, Who, When, and Why”, and shows them from different viewpoints such as planners, designers, and builders. This ensures that the organization’s architecture is viewed from all angles and no important aspect is missed. (Zachman J., 2022)

2. TOGAF (The Open Group Architecture Framework)

TOGAF provides a step-by-step approach called the Architecture Development Method (ADM), which guides organisations through stages such as defining architectural vision, designing business and IT architecture, planning implementation and management of continuous improvement. This method helps to ensure that the architecture work is not only well-structured, but also practical and consistent with business goals. (David N., 2025)

2.2.2 Review of Previous Studies (University Enterprise Architecture)

Enterprise Architecture (EA) has established itself as a crucial tool in higher education to integrate all components of the organisation from IT systems to academic programs in such a way that they are effectively aligned with the objectives of the institution. In order to improve the organisation of its information systems, for example, the Universitas Nahdlatul Ulama Surabaya in Indonesia implemented the TOGAF framework. This change accelerated the workflows at the university and improved the exchange of data between departments. Another example is a university in Europe whose use of EA to drive digital transformation has strengthened the institution's ability to adapt to new trends in education and service delivery.

Still, many universities face challenges in implementing Enterprise Architecture (EA). Changes sometimes encounter many problems or it can be difficult to get everyone in the institution to get involved on board. In addition, these frameworks, often developed for the business sector, may appear rigid or inappropriate in academic cultures. However, as universities adapt to their own context and receive the support of a strong management level, the benefits become clear for better system integration, more flexible processes and progress in achieving their strategic goals.

2.3 Related Sub-Systems

This section highlights the subsystems from UTHM's core functions such as finance and human resources. Each subsystem plays a crucial role in the UTHM's educational system.

2.3.1 Student Information System (SIS)

A Student Information System (SIS) is a software platform used by educational institutions to store, manage and process all student-related data during their time as a student. Academic information systems are one of the efforts of educational institutions to provide quality services to ensure convenience and comfort to students (Saputuri & Mulyani, 2022). Because of the role it plays, this system is a crucial module within the larger university ERP system. That is why it is important that its design is thoroughly planned out.

Vincent-Lancrin (2023) notes that in order for student information systems to support more relevant and innovative uses of education data, which show a realistic view of students experiences, the system has to have these three basic features: connect student-level data, which was collected during different time periods, maintain extensive information about students and facilitate the access of this data through different up-to-date reporting, visualisation and analysis tools. If this criteria is met, the system not only ensures that the institutional goals and projections based on data are relevant, it also makes the system as a whole more student-focused, providing better support and meeting student needs more effectively.

2.3.2 Financial System

This system is represented by UTHM's ePayment and other financial modules that manage all the financial transactions in UTHM. This includes student fees, employee salaries, procurement and financial reporting. It directly maps to the "finance" module of a comprehensive enterprise resource planning (ERP) system.

Previously, financial systems such as ePayment, payroll and procurement modules were usually built on **monolithic or tightly-coupled legacy architectures**. These old systems can effectively handle core financial tasks, but they are difficult to update, the integration with new services is slow, and the maintenance cost is high. Because each function, such as fee collection, payroll processing or procurement, is developed as a separate component

and interoperability is limited, the system lacks flexibility and real-time data sharing. This reflects the challenges described in the legacy financial system literature. This outdated architecture limits the scalability and needs modernisation to support integrated ERP-style financial management. (Sreenivasulu G., 2025)

2.3.3 Human Resource (HR) System

The Human Resources (SDM) system previously used by the University of Tun Hussein Onn Malaysia (UTHM) runs on an outdated independent platform. Although the old system can perform some basic human resources functions, such as recording attendance and storing employee information, most of its processes are isolated from each other. Therefore, tasks such as updating employee information or preparing salary statements require manual data transmission and repeated input. This not only reduces work efficiency but also increases the probability of mistakes.

In view of these limitations, UTHM decided to upgrade to a more modern and comprehensive integrated human resources management system. This upgrade has contributed to the launch of the "SMART HRM" platform, which integrates daily human resources processes into an interconnected environment. The new system can be connected to other university databases, support remote access by employees and managers, and provide functions such as online leave application, self-service personal information update portal and real-time performance monitoring.

Through these improvements, the human resources activities of various departments can share information more easily, operate more efficiently, and avoid delays caused by old-style monomer systems. The adoption of SMART HRM is also in line with the global trend of human resources digitalisation, ensuring that UTHM's human resources management can support the long-term strategic needs of universities.

2.3.4 E-Learning System

University Tun Hussein Onn Malaysia (UTHM) has an online learning system to support online and hybrid learning initiatives of the university. This online learning system is also known as Author UTHM, and it is mainly used for course delivery, evaluation, and as a medium to connect students and lecturers.

Previously, UTHM's online education relied on many different platforms, and teachers shared information through social network groups, Google Classroom, email, and other media. Due to the lack of integration of these tools with the university's main databases, there are problems such as inaccurate records, difficulties with assessment management, and challenges in monitoring student participation. As digital learning has been normalised in higher education, therefore, an integrated and structured system is needed for UTHM.

Author UTHM has overcome these limitations by creating a unified learning management system (LMS) that integrates functions such as course content, homework submission, tests, and teacher-student interaction. The platform supports automatic marking, real-time updates, and class materials, including notes and videos. The integrated login system allows users to access resources and submit homework easily, and also reduces the administrative workload of faculty and staff.

2.4 Technology Used

In order to support daily operations, the enterprise systems used by universities rely on a variety of software technologies, architectures and infrastructure components. These technologies ensure that each subsystem is efficient, safe, scalable, and can be integrated with other university functions.

2.4.1 SAP S/4HANA

SAP S/4HANA acts as the main "digital core" of the enterprise architecture that integrates the key operational functions of the university into a single data source to improve operational efficiency and reduce redundancy (Faisel M, 2019). The Student Life Cycle Management (SLcM) module manages students' journeys and integrates directly with SAP Finance to ensure real-time updates of fee payments without manual reconciliation (L. Qian, 2015). In addition, the SAP human capital management (HCM) module has modernized HRMIS 2.0, which can achieve accurate payroll processing, eliminate data redundancy and provide real-time analysis for strategic decision-making (Sarferaz S., 2024).

2.4.2 SAP Business Technology Platform (BTP)

SAP Business Technology Platform (BTP) is an integrated layer that connects core ERP and external systems. The SAP Integration Suite connects the SAP S/4HANA with the university's E-Learning platforms (AUTHOR LMS) to automatically synchronize course registration data between the registrar and the LMS (Li-Sen Z, 2024). In addition, BTP powers the "CENTRAL" portal by providing a secure single Sign-On (SSO) environment, allowing unified access to all subsystems (SMAP, ePayment, HRMIS) to improve security and user experience (Carretero J,2018).

2.5 Summary

The reviewed literature shows that by adopting Enterprise Architecture, institutions can significantly simplify and accelerate their day-to-day operations by organizing their workloads and integrating technology, processes and resources into a single unified system. The use of EA frameworks shows to further help organizations to effectively manage the architecture in practice. Additionally, the real-life example of EA being successfully implemented in a university in Indonesia illustrates the possibilities of successful system adaptation in the higher education sector. In this case, EA not only improves process efficiency but also allows systems to be flexible and customizable to specific institutional needs. Moreover, research on UTHM operations highlights the importance of up-to-date and practical sub-systems as they play a crucial part in effective and accessible institutional systems.

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